



Accelerate Test Execution with Adaptive Test

Reference: RA_453

Unlock Smarter Testing with DAS

Welcome to the Adaptive Test DAS Demo — a powerful illustration of how test programs can be made smarter, more flexible, and responsive to real-time conditions. Whether you're running it locally or envisioning a scaled deployment on remote servers, this demo is built to showcase adaptability.

What This Demo Demonstrates

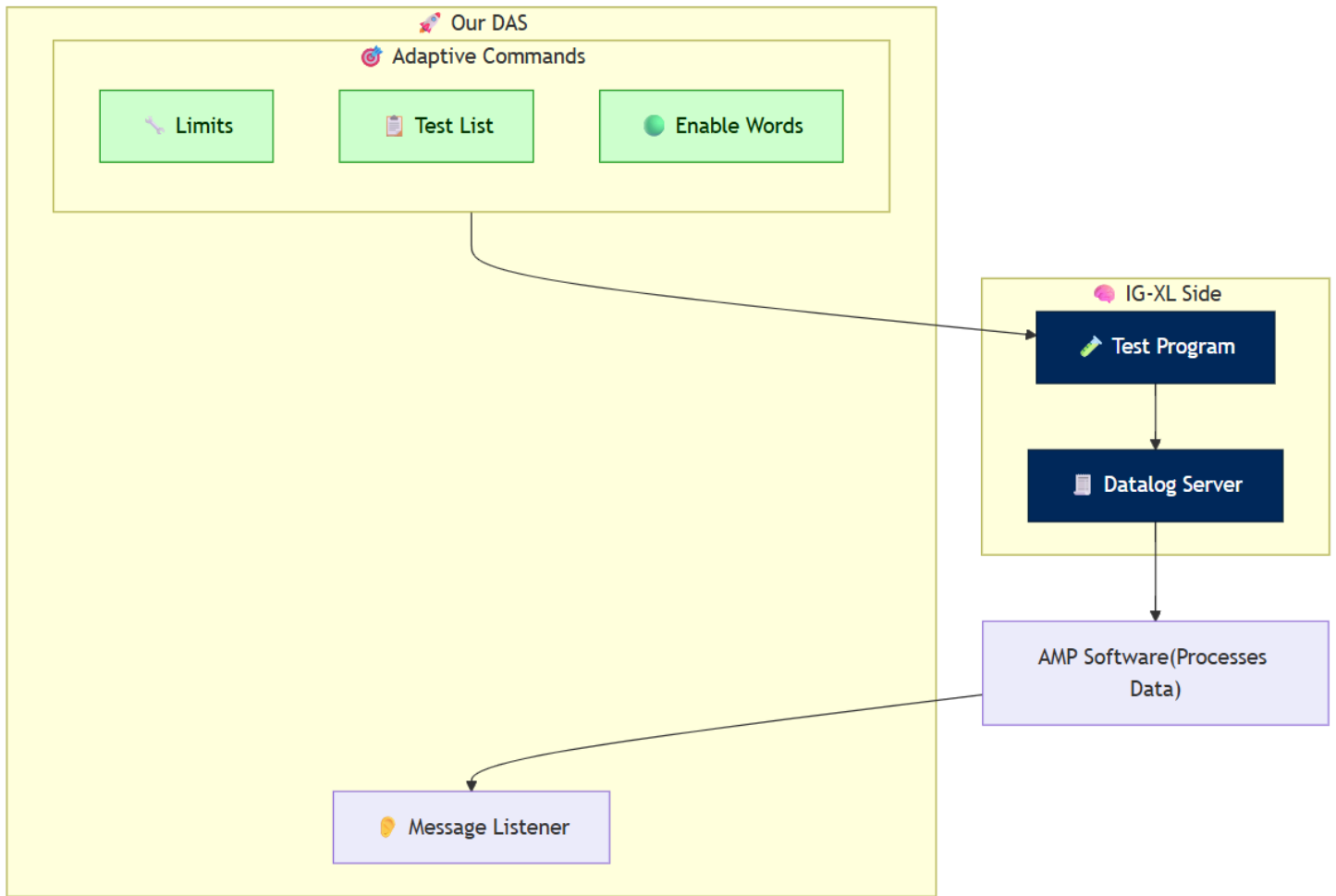
This demo highlights three key capabilities of our Data Analytics Solution (DAS):

1. **Enabling or disabling specific tests** — streamline test execution based on context or priorities
2. **Modifying test limits dynamically** — improve accuracy and adapt to production variability
3. **Controlling enable words** — activate or deactivate sub-flows conditionally

Additionally, the system offers robust observability:

- Monitor messages sent by the tester in real time
- Understand communication patterns driven by adaptive logic

Software Overview

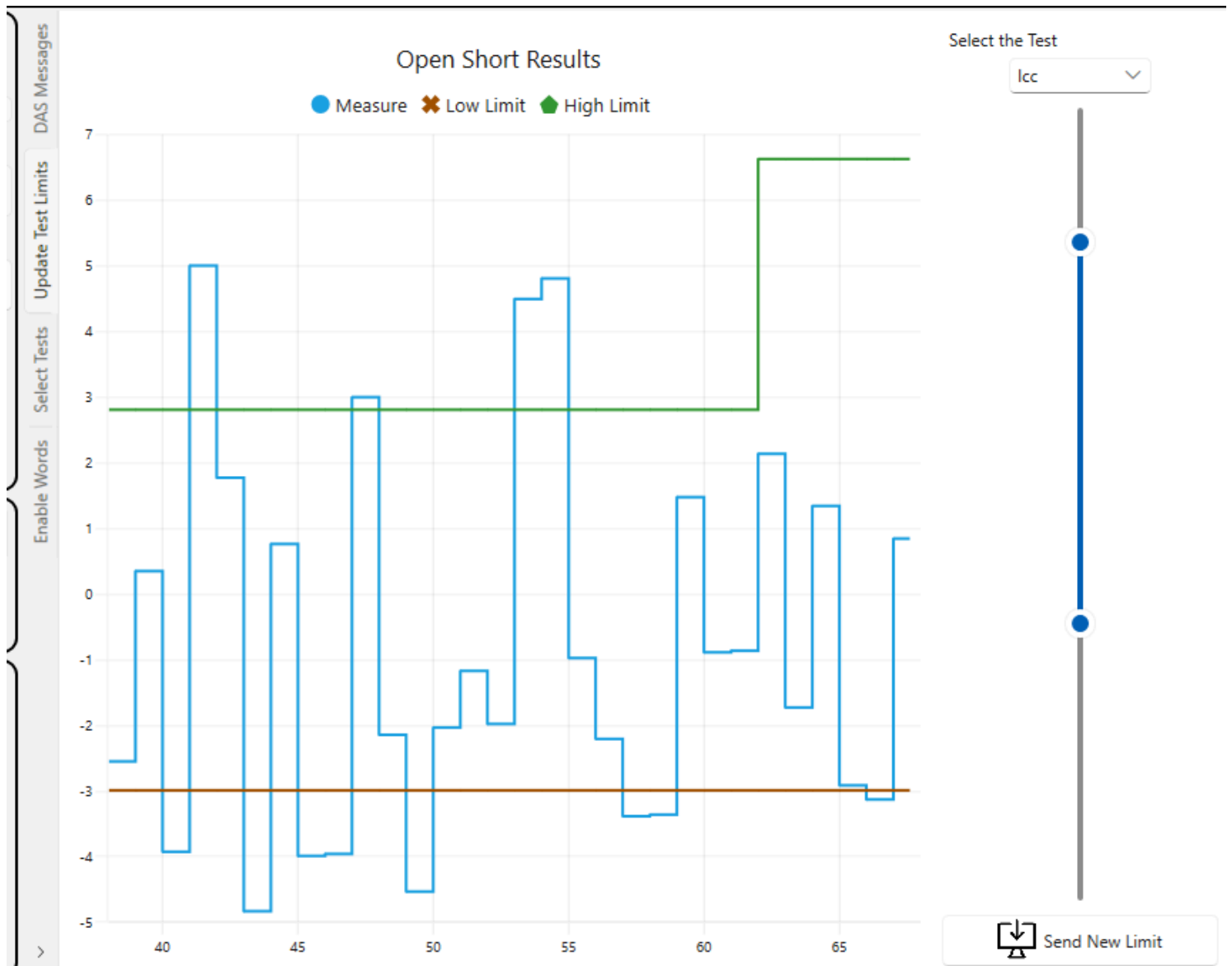


Explore the Demo Software

The demo application includes four dedicated sections:

DAS Section

View all messages generated by the tester. Clicking on a message reveals its detailed contents.



Enable Word Section

The test program includes two enable words that control sub-flow activation. Enable or disable them directly from this interface.

Component	Version
Framework	.NET 8
Environment	Windows 10
Excel	2016 (64-bit)
IGXL	10.60.02

Extend the Demo to Your Environment

To implement a similar system, we recommend reviewing the following guides:

Reference Architecture	Description
RA_0468	How to implement a DAS in C#
RA_0478	How to send an adaptive command in C#

See It in Action

Explore the complete [Instructions to Run the Demo](#) or browse the image gallery to discover key functionalities.

Looking to integrate this approach into your test strategy? [Contact our team](#) to explore how DAS can support your roadmap.

This demo is part of our Reference Architecture Series – built to inspire smarter test development.



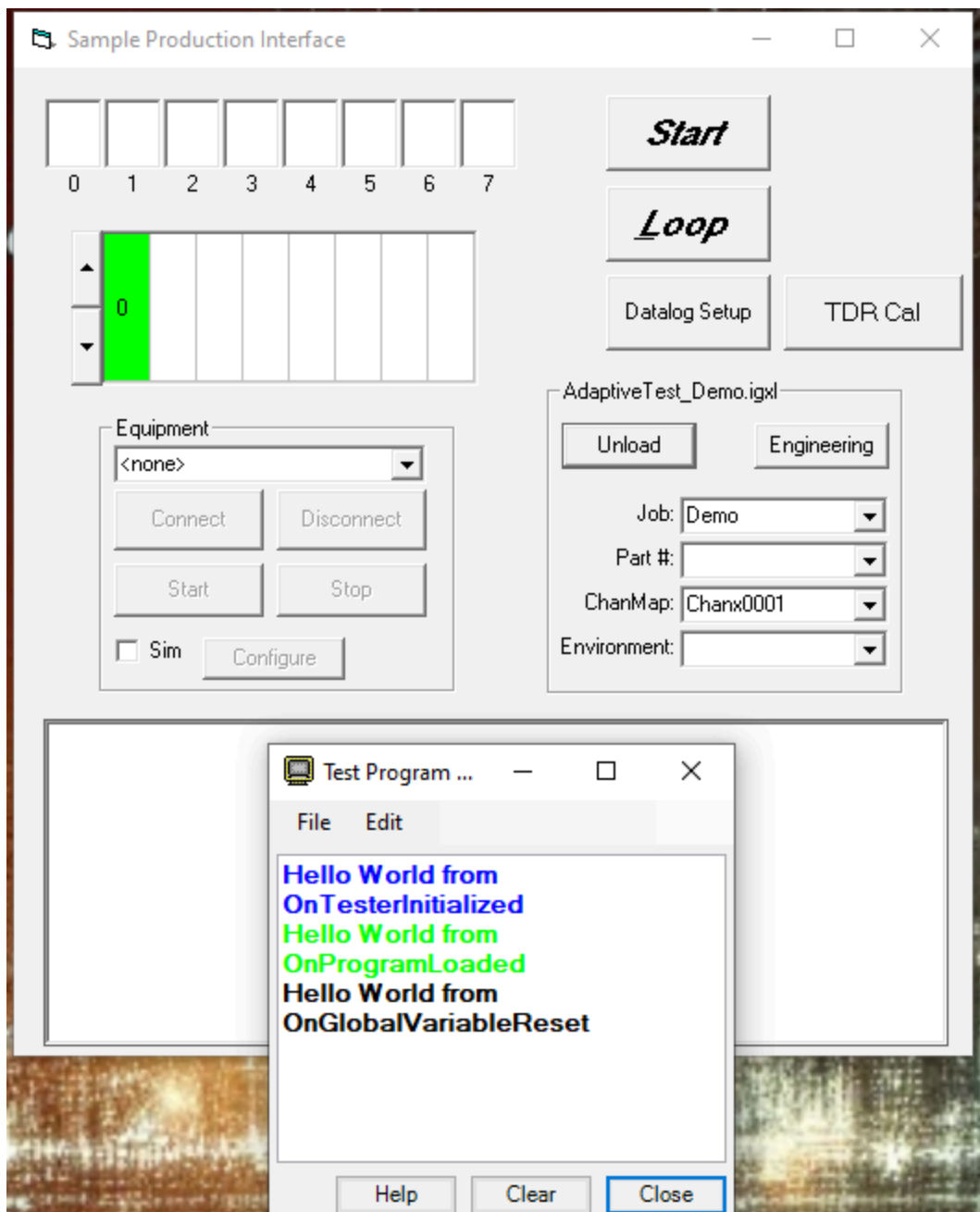
How to Run the Adaptive Test Demo

Reference: RA_453

This section provides step-by-step instructions to explore the key features of the Adaptive Test DAS Demo.

Step 1: Load the Test Program

Use the SimpleOI interface to load your test program.



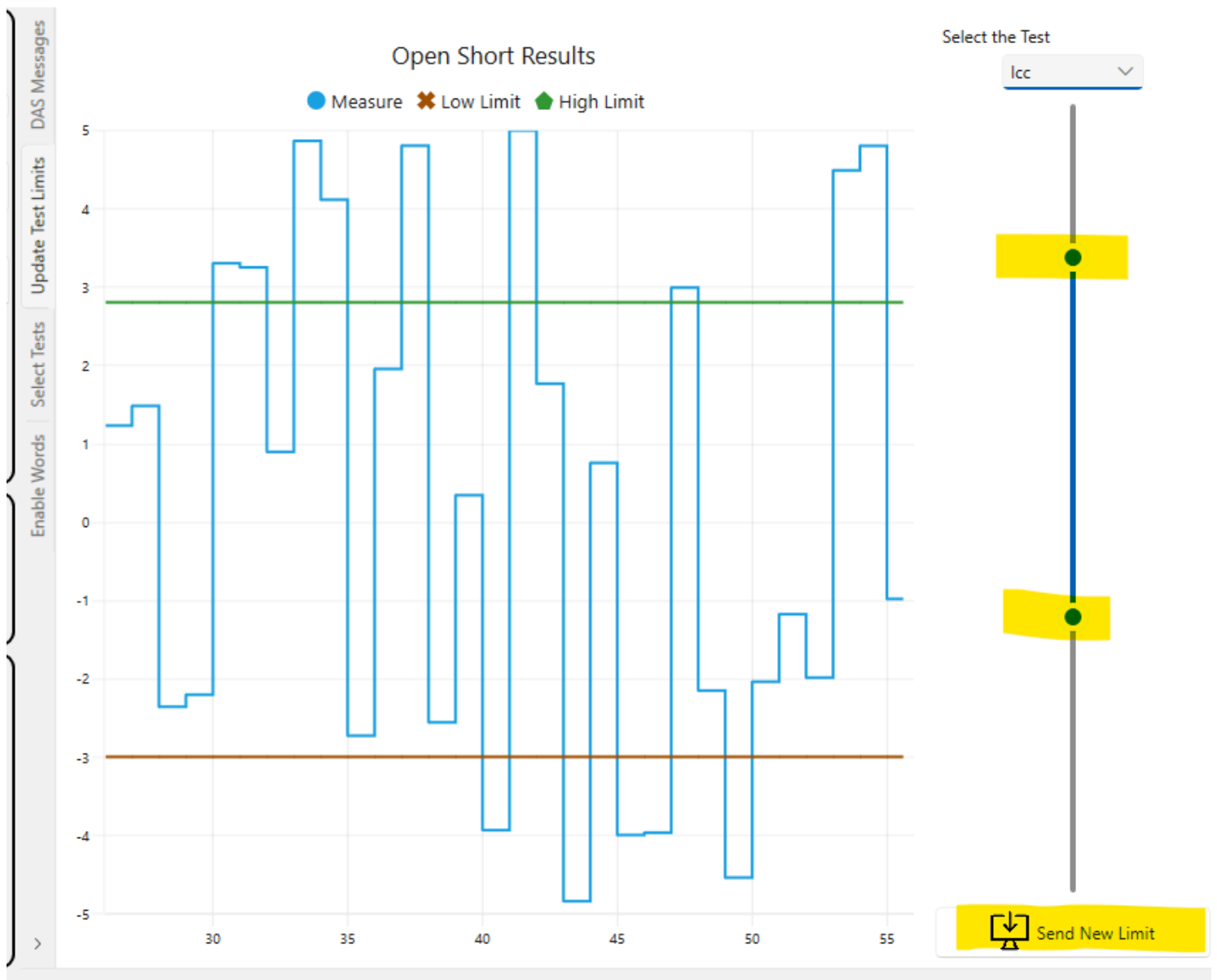
Step 2: Execute the Program in Loop Mode

Click the **Loop** button to execute the program 10 times. Once completed, press the **Stop** button.



Step 3: Adjust Test Limits Dynamically

In the demo application, click **Update Test Limits**. Use the slider to modify the thresholds, then click **Send Limits** to apply them.



Run the test a few times from the SimpleOI interface to observe the impact.

Step 4: Observe the Updated Limits

The graphical output will reflect the new limit values.

DIA

Adaptive Tests

TERADYNE

Communication Status

DAS URL

<http://127.0.0.1:3000/tems/>

Connect the DAS



Disconnect the DAS



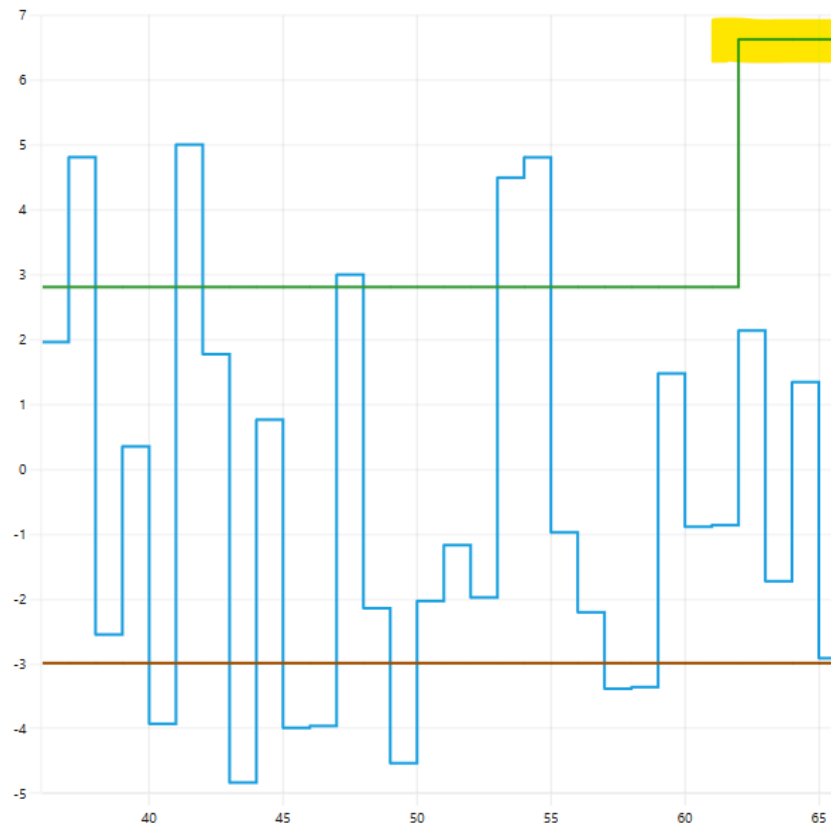
Clear Adaptive Command

Adaptive Command

DAS Messages
Update Test Limits
Select Tests
Enable Words

Open Short Results

● Measure ✕ Low Limit ◆ High Limit



Select the Test

lcc



Send New Limit

In DAS, select the **Adaptive_Change limits** message. Click on the message to inspect its contents.

Time Stamp	Test Cell	Name	Content
08:01:48.9137	TEMS-01	ADAPTIVE_CHANGE	TEP_ACTIONS": [{"ACTION":"UPDATE_LIMITS","REQUEST_ID":1,"TESTSTEP_NAMES": ["Icc"],"LIMITS":{"LIMIT_NAMES":

```
{
  "TIME_STAMP": "2025-05-05T08:01:48.9137009Z",
  "LOT_ID": "UNKNOWNLOT_PID_8380_1",
  "SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
  "DAS_ID": 1,
  "TESTSTEP_ACTIONS": [
    {
      "ACTION": "UPDATE_LIMITS",
      "REQUEST_ID": 1,
      "TESTSTEP_NAMES": [
        "Icc"
      ],
      "LIMITS": {
        "LIMIT_NAMES": [
          "Icc"
        ]
      }
    }
  ]
}
```

You can also observe the received message in the IGXL test program output.

Teradyne_IG-XL_DataCollect_8380 - OutputWin
File Edit Collect View Help

Site
Sort
Bin

0
1
1

OnProgramEnded: 2025-05-05 10:01:40.390
Adaptive Change: The limits {Icc} on test step {Icc} were changed. New low limit = -3. New high limit = 6.61954453733064.
OnProgramStarted: 2025-05-05 10:01:48.910
Device#: 63

Number	Site	Test Name	Pin	Channel	Low	Measured	High	Force	Loc
<Icc_Fake>									
1000	0	Icc	-1		-3000.0000 mA	2127.3046 mA	6619.5445 mA	0.0000	0

Step 5: Work with Enable Words

This part demonstrates how to activate a sub-flow using enable words. Initially, only the test **ICC** is active.

To explore more:

- Access the engineering view in SimpleOI (password: Teradyne)
- Open the test program and locate the DemoFlow
- Visualize the flow and locate the enable word logic

AdaptiveTest_Demo.igxl [Book1] - Teradyne IG-XL DataTool - Offline Mode - C:\Users\admin\source\rep

File										Home		Insert		Page Layout		Formulas		Data		Review		View		IG-XL Main		IG-XL Tools		Data Analysis		Tell r	
																														Job: Demo	
Restart		Select Sheet		Previous Sheet		Apply Style		Formulas		Insert		Delete		Run Options		Calibrate TDR		Validate Job		Run		Output Window		Profiler Off		DataCollect Setup		Datalog On		Channel Map: Chanx00	
DataTool		Sheet																						Data Collect				Part:			
K14																															
1																															
2																															
1 2 A B C D E F G H I J																															
1 Flow Table TERADYNE																															
2																															
3																															
4																															
5																															
6																															
7																															
8																															
9																															
10																															
11																															
12																															

Gate					Command				
Label	Enable	Job	Part	Env	Opcode	Parameter		TName	TNum
	OnePin				Test	lcc_Fake			1000
	DualPin				call	OnePin_Flow			
					call	DualPin_Flow			
					set-device				
					stop				

Click on the OnePin EnableWord option and run the test program once.

Select the enable words you want to enable or disable

☒ Enable or Disable the enable word : OnePin

☐ Enable or Disable the enable word : DualPin

You'll see that more tests are now activated:

Adaptive Change: The following enable words were enabled: {OnePin}.

OnProgramStarted: 2025-05-05 10:09:38.800
Device#: 68

Number	Site	Test Name	Pin	Channel	Low	Measured	High	Force	Loc
<Icc_Fake>									
1000	0	Icc	-1		-3000.0000 mA	833.5900 mA	6619.5445 mA	0.0000	0
OnePin									
<OnePin_NoHighLimit>									
200	0	OnePin_NoHighLi	-1		100.0000 m	1.1235	N/A	0.0000	0
<OnePin_NoLowLimit>									
300	0	OnePin_NoLowLim	-1		N/A	500.0000 m	900.0000 m	0.0000	0
<OnePin_NoLimit>									
400	0	OnePin_NoLimit	-1		N/A	-700.0000 m	N/A	0.0000	0
<OnePin_MilliAmpUnit>									
500	0	OnePin_MilliAmp	-1		-1000000000.0000 nA	17500000000.0000 nA	(F) 0.0000 nA	0.0000	0

A new Adaptive Change message will also be generated:

08:09:38.8061	TEMS-01	ADAPTIVE_CHANGE	ID_8380_1","SUBLOT_ID":"UNKNOWNSUBLOT_PID_8380_1","DAS_ID":1,"TESTSTEP_ACTIONS":[],"ENABLEWORD_ACTIONS":[{"ACTION":"ENABLE","REQUEST_ID":2,"ENABLEWORDS":
---------------	---------	-----------------	---

```

"TIME_STAMP": "2025-05-05T08:09:38.806154Z",
"LOT_ID": "UNKNOWNLOT_PID_8380_1",
"SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
"DAS_ID": 1,
"TESTSTEP_ACTIONS": [],
"ENABLEWORD_ACTIONS": [
  {
    "ACTION": "ENABLE",
    "REQUEST_ID": 2,
    "ENABLEWORDS": [
      "OnePin"
    ],
    "STATUS": "OK"
  }
]

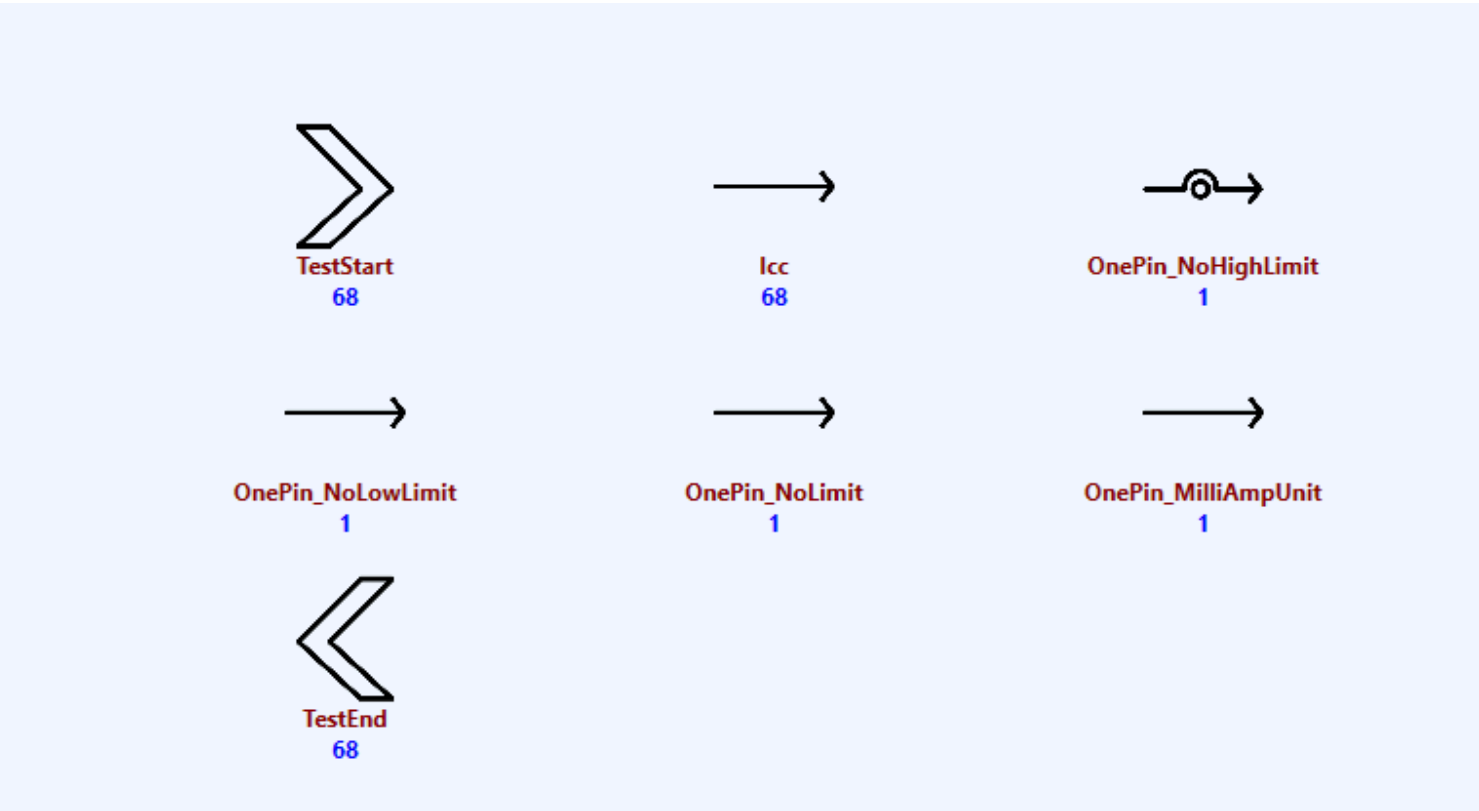
```

Step 6: Enable or Disable Tests

This step shows how individual tests can be selectively enabled or disabled.

- Go to the **Select Test** tab in the DAS interface
- A count of executions for each test is shown in blue
- For best demo results, activate **DoAll** in IGXL Run Options

Now, disable the `OnePin_NoHighLimit` test by selecting it.



Run the test program once. You'll see that the test is skipped.

08:25:20.0344	TEMS-01	ADAPTIVE_CHANGE	ID_8380_1";SUBLOT_ID":"UNKNOWN";SUBLOT_PID_8380_1";DAS_ID":1,"TESTSTEP_ACTIONS":[{"ACTION":"DISABLE","REQUEST_ID":3,"TESTSTEP_NAMES":
<pre>{ "TIME_STAMP": "2025-05-05T08:25:20.034446Z", "LOT_ID": "UNKNOWNLOT_PID_8380_1", "SUBLOT_ID": "UNKNOWN";SUBLOT_PID_8380_1", "DAS_ID": 1, "TESTSTEP_ACTIONS": [{ "ACTION": "DISABLE", "REQUEST_ID": 3, "TESTSTEP_NAMES": ["OnePin_NoHighLimit"] }], "STATUS": "OK" }</pre>			

You can confirm this in IGXL:

OnProgramEnded: 2025-05-05 10:09:38.870
Adaptive Change: The following test steps were disabled: {OnePin_NoHighLimit}.
OnProgramStarted: 2025-05-05 10:25:20.030
Device#: 69

Number	Site	Test Name	Pin	Channel	Low	Measured	High	Force	Loc
<Icc_Fake>									
1000	0	Icc	-1		-3000.0000 mA	-4192.8535 mA (F)	6619.5445 mA	0.0000	0
OnePin									
<OnePin_NoLowLimit>									
300	0	OnePin_NoLowLim	-1		N/A	500.0000 m	900.0000 m	0.0000	0
<OnePin_NoLimit>									
400	0	OnePin_NoLimit	-1		N/A	-700.0000 m	N/A	0.0000	0
<OnePin_MilliAmpUnit>									
500	0	OnePin_MilliAmp	-1		-1000000000.0000 nA	17500000000.0000 nA (F)	0.0000 nA	0.0000	0

Download the Demo

[Download the demo from here](#)